

CSC 3320: System-Level Programming



Review for the Final Exam

Instructor: Dr. Md Mahfuzur Rahman
Department of Computer Science, GSU

Announcements

- Final Exam date announced earlier on iCollege
 - (Note your time from iCollege announcement)
 - Note the final exam has a different schedule.
 - No make-ups!

Disclaimer

This review merely highlights some of the important areas that you may focus for the final exam. Samples shown in this review are just few examples (not exhaustive in terms of either examples or topics). For complete preparation, please refer to all weekly slides/books/lecture notes for the chapters mentioned in the exam syllabus slide.

Question Types and Distributions

- MCQ (~40%)
 - May involve coding questions (specifically output determination)
 - Other conceptual questions
- Short Answer type (40-50%)
 - Piece of code writing, output determination w/o options, comprehension questions
- Write/Fill in small piece of code or small programs. (10-20%)
 - Fill-in codes
 - Write small piece of programs

Exam Syllabus

- The following chapters will cover 50% of the final exam points
 - Chapter 12 (Pointers and Arrays)
 - Chapter 13 (Strings)
 - Chapter 16 (Structures, Unions, and Enumerations)
 - Chapter 17 (Dynamic Memory Allocation)
 - Chapter 13 (Unix Book: File Management, Process Management using system calls)
- The earlier chapters/topics will make up around 50% of the final exam.

A general suggestion

Visit/learn from:

- All the HW assignments
- Exam 1
- Exam 2

They will be a very good source to understand example questions and the difficulty you may expect in the final.

Understand the following topics

- Regular expression question
- File permission question?
- Variable naming rules/static variables
- Increment/decrement operators, logical and relational operators, precedence
- Continue/break/switch statement
- Loops, arrays, functions
- Call by value, call by reference
- Enumeration constants (HW5)
- Linked list manipulation (Delete/insert/search first/Last/some node)
- Simple recursion on linked list
- String printing/manipulation (ASCII manipulation)
- Structure/Union: learn to calculate sizes of a union/structure (HW5)
- Dynamic memory allocation: malloc and free
- File management system calls and flags: open, read, write()
- Process creation (specially nature of fork system call)
- Precedence of operators in `*++p`, `++*p`, `(*p)++`, `*p++`

(Note: this list is not exhaustive. Also follow lectures/slides)

Practice Regular Expression

Write regular expressions for following patterns.

- Phone number with the format ###-###-####
- Date with the format MM/DD/YYYY
- Passwords: strings with at least one upper case letter, one lower case letter, and one digit

Choose all matched strings of the given extended regular expression.

- (1) a(bc)+de
A) abcde B) ade C) abcbcede D) abc
- (2) [a-z\s]*hello
A) Othello B) hello C) 2hello D) say hello

Change File Permissions

Another example with octal numbers

```
mweeks@carmaux:~$ ls -l example.txt
-rw-r--r-- 1 mweeks mweeks 51 2007-08-23 16:06
example.txt
mweeks@carmaux:~$ chmod 765 example.txt
mweeks@carmaux:~$ ls -l example.txt
-rwxrw-r-x 1 mweeks mweeks 51 2007-08-23 16:06
example.txt
```

Notice

- 7 = 111 (binary) r “on”, w “on”, x “on”
- 6 = 110 (binary) r “on”, w “on”, x “off”
- 5 = 101 (binary) r “on”, w “off”, x “on”

Review from Exam 2

What will happen if you run the program? Why?

```
#include<stdio.h>
int main()
{
    int n;
    for (n = 9; n!=0; n--)
        printf("n = %d", n--);
    return 0;
}
```

Review from Exam 2

What will be the output of the following program?

```
#include<stdio.h>
void f(int *p, int *q)
{
    q = p;
    *p = 15;
}
int i = 0, j = 1;
int main()
{
    f(&i, &j);
    printf("%d %d", i, j);
    getchar();
    return 0;
}
```

Ans: 15 1

Finding the largest element

```
int find_largest(int *p, int n)
{
    int i;
    int max_value = -100; // Assume any very small number
    for(i = 0; i < n; i++)
    {
        if (p[i] > max_value)
            max_value = p[i];

        /* The following two lines should also work */
        // if(*(p+i) > max_value)
        //  max_value= *(p+i);
    }

    return max_value;
}
```

What will be the output of the following program?

```
#include <stdio.h>
int main()
{
    int i = 0;
    for (i=1; i<20; i++)
    {
        switch(i)
        {
            case 0:
                i += 5;
            case 1:
                i += 2;
            case 5:
                i += 5;
            default:
                i += 4;
                break;
        }
        printf("%d ", i);
    }
    return 0;
}
```

Output: 12 17 22

What will be printed?

```
#include<stdio.h>
int main()
{
int a1 = 9, a2 = 10, a3 = 3;
if (a1 < a2 > a3)
    printf("TRUE");
else
    printf("FALSE");
return 0;
}
```

You might think this means: $a1 < a2$ AND $a2 > a3$
This means:
 $(a1 < a2) > a3$

Output: FALSE

Process creation

```
#include <stdio.h>
#include <unistd.h>
int main() {
    printf("Before fork()\n");

    pid_t pid = fork();
    if (pid == 0) {
        // Code executed by the child process
        printf("Child process: PID = %d\n", getpid());
    }
    else if (pid > 0) {
        // Code executed by the parent process
        printf("Parent PID = %d, Child PID = %d\n", getpid(), pid);
    }
    else {
        // fork() failed
        perror("fork failed");
    }
    printf("This line is executed by both parent and child processes\n");
    return 0;
}
```

Process creation

How many times "Hello World" will be printed? How many processes will be created altogether?

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    fork();

    printf("Hello World\n");

    return 0;
}
```

Output: 4 times

Increment/Decrement Operators

```
int main ( )  
{  
    int a, b, c, d;  
    a=15; b=10;  
    c =++a - b;  
    d=b++ +a;  
}
```

What will be the value of a, b, c, and d?

Output: a = 16, b = 11, c = 6, d = 26

Increment/Decrement ops

What will be the value of x and y after the execution of the following statements?

```
int i=6;  
int j=4;  
int y;  
y=(j++) * (++i) ;
```

What will be the value of i, j, and y?

Output: i = 7, j = 5, and y = 28

Logical operators

```
#include <stdio.h>
#include<stdio.h>
int main()
{
    int a = 1;
    int b = 1;
    int c;
    c = (b == (a += b)) ;
    printf("%d", a);
    return 0;
}
```

What will be the output?

Output: a = 2, and c = 0

Review from Exam 2

```
#include<stdio.h>
int main()
{
    int i = -2;
    while (i <= 5)
    {
        if (i >= 0)
            break;
        else
        {
            i++;
            continue;
        }
        printf("Assembly language Programming");
    }
    return 0;
}
```

Ans: 0

Continue/break statement

```
int main( )
{
    int i, j, sum=0;
    for(i=1; i<=10; ++i)
    {
        if(i%2==0)
            continue;
        for(j=1; j<=i; j++)
        {
            sum+=j;
            break;
        }
    }
}
```

What will be the value of `sum`?

Output: `sum = 5`

ASCII values

What will be the value of x after execution of the following statements?

```
int x, y=10;  
char z='2';  
x=y+z;
```

Ans: x = 60

Pointers

```
int main( )  
{  
    int m=10, n, *b;  
    b=&m;  
    n=*b;  
    m=*b+n;  
}
```

What will be the value of m and n?

Ans: m = 20 and n = 10

Pointers and Arrays

```
#include < stdio.h >
int main( )
{
    int arr[4][5];
    int i, j;
    for (i=0; i<4; i++)
    {
        for (j=0; j<5; j++)
        {
            arr[i][j] = 10 * i + j;
        }
    }
    printf("%d", *(arr[1]+9));
    return 0;
}
```

Learn what arr, arr[0], arr[1],
*arr, *(*arr) mean

What will be printed on the screen?

Ans: 24

String Manipulation

```
int main()
{
    char str[] = "GeorgiaState";
    printf("%s", str + str[3] - str[2]);
    return 0;
}
```

What will be the output?

Ans: rgiaState

Study HW4 to practice string problems

- Also learn other string functions: strcmp, strcat, strcpy, strlen, etc.

Structure/Union

See problems from HW5

Reminder

- This is just a few examples. Go back to the topics list and read from weekly slides, notes, HW1-HW5, and other sources.